File 31: UGA Intern Workstream — Page 4: USACE Savannah District 7-Year Monitoring & Release Schedule

> ***Ecology & Biological Monitoring Training Manual Target School***: *University of Georgia (UGA) Odum School of Ecology / Warnell School of Forestry* ***Project Area***: *Upper Savannah Basin & Chattahoochee River Spawning Headwaters* ***Supervising Board***: *Hunter Morris (Co-Founder & Managing Director) & Hadi Irvani (Board Member)*

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1. Regulatory Context: Credit Release Milestones

A mitigation bank does not receive all of its commercial stream credits at once. The **USACE Savannah District** releases credits in stages over a **7-year monitoring window**. These releases are strictly contingent upon demonstrating physical channel stability and biological success.

As a UGA Intern, you will compile the data from your biological, thermal, and vegetative surveys into our annual **Ecological Monitoring Reports**. The USACE will not release credit reserves if we fail to submit these reports on time, or if our biological metrics fail to meet target thresholds.

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2. The 7-Year Stream Credit Release Schedule

Our flagship bank at Anderson Creek yields exactly **12,900 credits**. The phased release schedule and required biological performance thresholds are detailed below:

Year	Released %	Released Credits	Critical Ecological & Geomorphic Performance Milestones
Year 0	15%	1,935 credits	**Baseline**: Mitigation Banking Instrument (MBI) approved; conservation easement recorded; credit security bonds posted.
Year 1	15%	1,935 credits	As-built survey submitted; bank stability index (BEHI) rated "Good" or better; vegetative survival $\ge 90\%$.
Year 2	15%	1,935 credits	Zero stream bank failure after bankfull storm event; vegetative density $\ge 260\text{ stems/acre}$; invasive coverage $<10\%$.

Year	Released %	Released Credits	Critical Ecological & Geomorphic Performance Milestones
Year 3	10%	1,290 credits	Stream temperature loggers prove max summer temperature $<65^\circ\text{F}$; benthic macroinvertebrate EPT index increases by $\ge 15\%$.
Year 4	10%	1,290 credits	Rosgen cross-section width-to-depth ratios stable ($\pm 10\%$ of as-built designs); native vegetation canopy covers $\ge 40\%$ of plot surface.
Year 5	15%	1,935 credits	Benthic EPT index matching reference stream conditions; native vegetative density stable; invasive coverage $<5\%$.
Year 6	10%	1,290 credits	Multi-year thermal logging proves consistent temperature suppression; biological community shows high ecological lift.
Year 7	10%	1,290 credits	**Final Release**: Complete geomorphic stability verified; permanent stewardship transferred to Georgia Land Trust.

3. Calculating the Benthic EPT Index

To verify biological recovery at Year 3 and Year 5, you must calculate the **EPT (Ephemeroptera, Plecoptera, Trichoptera) Index**. EPT organisms are highly sensitive to siltation and organic pollutants; their abundance is direct evidence of water quality improvement.

$$\text{EPT Index} = \frac{\text{Taxa Count of (Mayflies + Stoneflies + Caddisflies)}}{\text{Total Taxa Count of All Collected Macroinvertebrates}} \times 100\%$$

Worked Example:

You collect a sample at Anderson Creek Station 02 containing the following macroinvertebrates:

- **Ephemeroptera (Mayflies)**: 4 distinct taxa (Heptageniidae, Baetidae, Ephemerellidae, Siphlonuridae)
- **Plecoptera (Stoneflies)**: 2 distinct taxa (Perlidae, Pteronarcyidae)
- **Trichoptera (Caddisflies)**: 3 distinct taxa (Hydropsychidae, Limnephilidae, Rhyacophilidae)

- **Chironomidae (Non-biting Midges - Pollution Tolerant)**: 5 distinct taxa
- **Odonata (Dragonflies - Moderately Tolerant)**: 2 distinct taxa

1. Calculate EPT Taxa:
\text{EPT Taxa} = 4 + 2 + 3 = 9\text{ taxa}
2. Calculate Total Taxa:
\text{Total Taxa} = 9 + 5 + 2 = 16\text{ taxa}
3. Calculate EPT Index:
\text{EPT Index} = \frac{9}{16} \times 100\% = 56.25\%
Ecological Assessment: An EPT Index ***>50\%*** represents an **"Excellent"** water quality rating, satisfying the USACE Year 3 ecological lift requirement.

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4. USACE Annual Submission Checklist

Before submitting the annual report to the Savannah District IRT, you must verify that the following files are compiled and attached:

- [ ] **Executive Summary Sheet**: Formatted with Co-Founder Hunter Morris's signature and Board Member Hadi Irvani's review stamp.
- [ ] **HOBO Logger Data Plots**: Continuous summer water temperature telemetry graphs with critical 65°F line highlighted.
- [ ] **Vegetation Plot Sheets**: Native stem counts, diameter measurements, and calculated stems-per-acre density for all 5 monitoring plots.
- [ ] **Macroinvertebrate Taxa Log**: Complete species list, calculated EPT indices, and photographic voucher plates of identified specimens.
- [ ] **Geomorphic Cross-Sections**: Overlay plots comparing pre-construction, as-built, and current year channel cross-sections to prove dimensional stability.
- [ ] **Ground Photographic Plates**: Fixed-point photos facing upstream, downstream, and perpendicular at every cross-section monument.